

STG17 TECHNICAL WORKSHOP · DAY 1 · 09:45–10:15

The AI Family Tree

How the concepts fit together — from AI to AGI, from large language models to agents and MCP

Model

AI

AGI

LLM

Prompt

RAG

Fine-tuning

Agents

MCP

AI-ready

African Development Bank · STATAFRIC (African Union)

Thirty minutes to share one language all week



1. Name

Define ten concepts: model, AI, AGI, LLM, prompt, RAG, fine-tuning, agent, MCP and AI-ready.



2. Place

Put each concept in one of three layers: the model's weights, its context, or action.



3. Assess

Say what each one can — and cannot — do for the production of official statistics.



The diagnostic question to take away

Faced with any proposal: which layer does it touch — the weights, the context, or the output turned into action?

Six steps, one common thread

01

2 min

Why a shared vocabulary

What is really new since 1976

02

6 min

The tree: from AI to AGI

What contains what, and where AGI sits

03

6 min

The model: what an LLM is

What a model is, and its limits

04

6 min

Steering: prompt, RAG, fine-tuning

Four places to intervene

05

5 min

Acting: agents, tools, MCP

When output becomes action

06

5 min

For official statistics

AI-ready, can / cannot, red line

SECTION 01

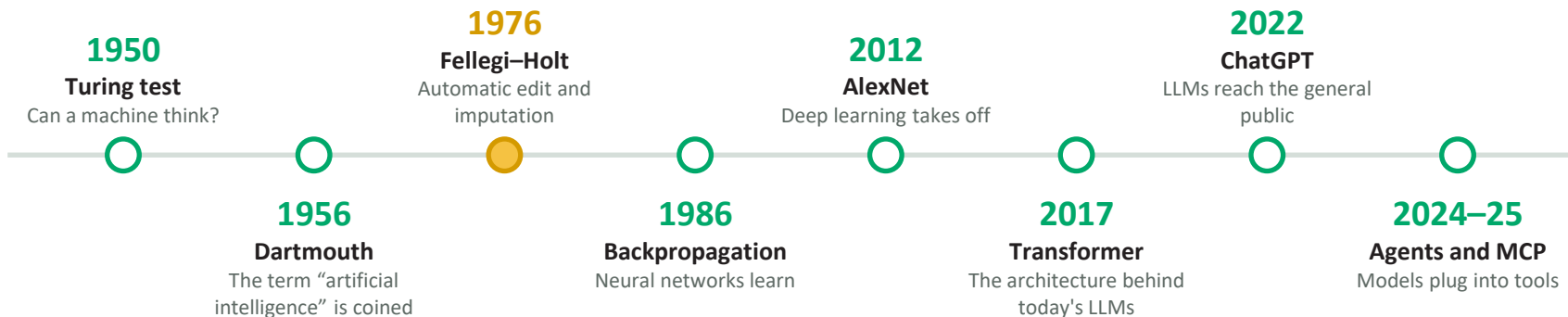
01

Why a shared vocabulary

If machines have been deciding since 1976, what is actually new?



Machine decision-making isn't new: your profession has done it since 1976



Fellegi and Holt (1976) already described a system that flags inconsistent responses, decides which fields to change and computes replacement values. **What is new is not that a machine decides: it is that one system generalises to a thousand tasks, directed in plain language.**

Fellegi, I. P. & Holt, D. (1976), JASA 71(353) · Rumelhart, Hinton & Williams (1986), Nature 323 · Vaswani et al. (2017), NeurIPS 30.

What has broken, and what has not



What remains

Machine decision-making

in your office since 1976

Fitting a function to data

the core of your discipline

The duty to reproduce

which none of these technologies changes

The need to evaluate

harder, never optional



What has broken

One system, a thousand tasks

including unforeseen ones

The interface

plain language, not code

Who can direct it

no programmer in between

The cost of a trial

near zero, so everyone tries



The trap: LLM, RAG, fine-tuning, agent and MCP are sold as rival products. They are layers of the same machine.

Framework: Bresnahan & Trajtenberg (1995), general purpose technologies · Exposure: Eloundou et al. (2024), Science 384.

SECTION 02

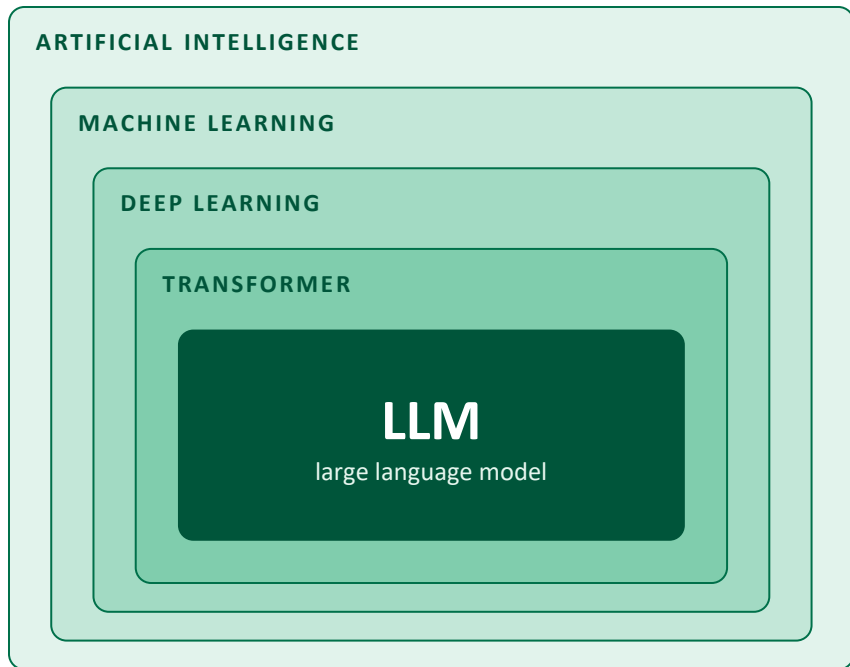
02

The tree: from AI to AGI

Is an LLM a kind of AI? And does AGI already exist?



Each concept sits inside the one before

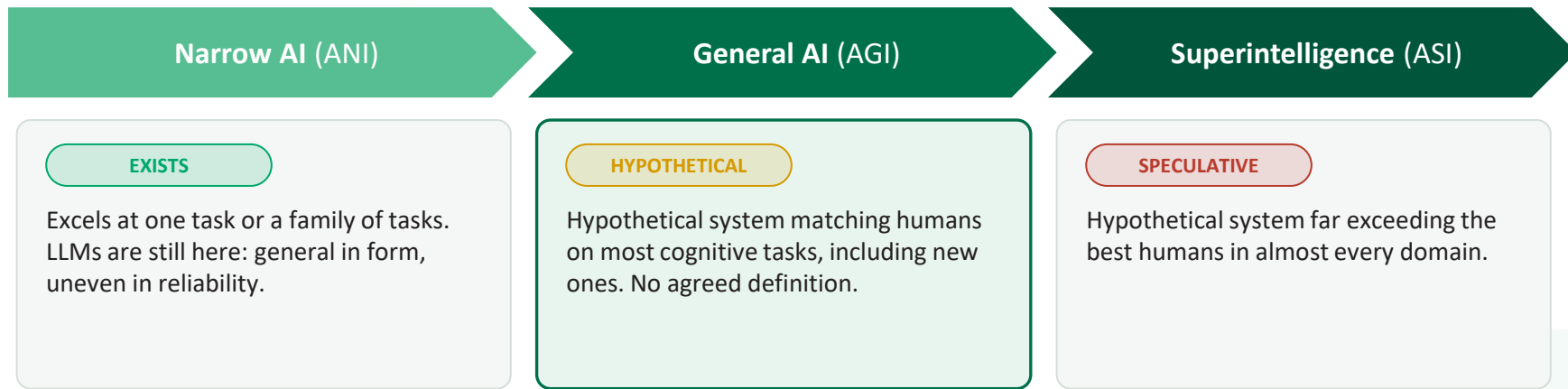


- **AI**
A machine performs a task considered “intelligent”, even using rules.
e.g. Fellegi–Holt editing rules
- **Machine learning**
Rules learned from examples rather than written by hand.
e.g. Automatic occupation coding
- **Deep learning**
Learning with many-layered neural networks.
e.g. Detecting buildings in satellite images
- **Transformer**
Architecture based on “attention” (2017) that reads text in parallel.
e.g. Machine translation FR ↔ EN
- **LLM**
A giant transformer trained to predict the next word.
e.g. First draft of a release commentary

A tender that asks for “artificial intelligence” has specified nothing: that ring also contains a rule-based system from 1976.

AGI: a contested horizon, not an available product

▼ *Today: versatile systems with a “jagged frontier”*



Vocabulary discipline: “generative AI” is not “AGI”. If an offer promises “AGI” or “near-AGI”, ask: which measured capability, on which task, today?

Morris et al. (2024), “Levels of AGI”, ICML · Blili-Hamelin et al. (2025), “Stop treating AGI as the north-star goal”, arXiv:2502.03689.

Three definitions of AGI, three different finish lines



By breadth

Match at least a skilled adult across a wide range of cognitive tasks, with graded levels.

Morris et al. (2024), Google DeepMind



By economic value

Highly autonomous systems that outperform humans at most economically valuable work.

OpenAI Charter (2018)



By learning

Acquire a genuinely new skill as efficiently as a human, from only a few examples.

Chollet (2019), ARC test

2047

FORECAST

median year by which 2,778 AI researchers put a 50% chance on “human-level” machines across all tasks.

Grace et al. (2024), “Thousands of AI Authors on the Future of AI”, arXiv:2401.02843 — individual estimates vary widely.



For a statistics office

Plan on capabilities measured today. AGI is not a layer you can buy, nor a system you can configure or audit.

SECTION 03

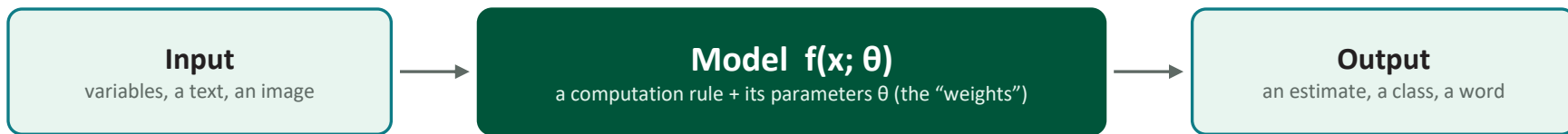
03

The model: what an LLM is

If a model writes well, does it actually know things?



What is a model? A function whose parameters are estimated



Training: estimate θ from data — once, expensive

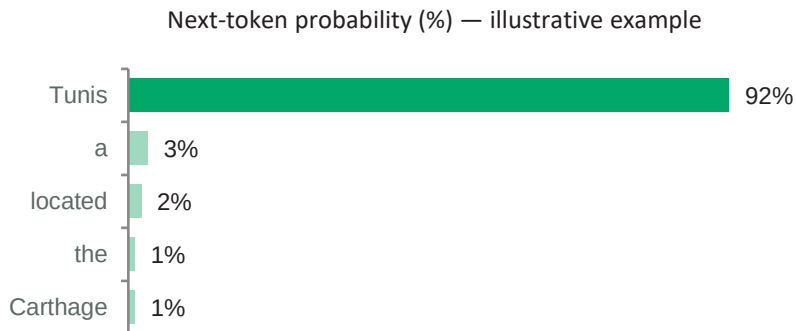
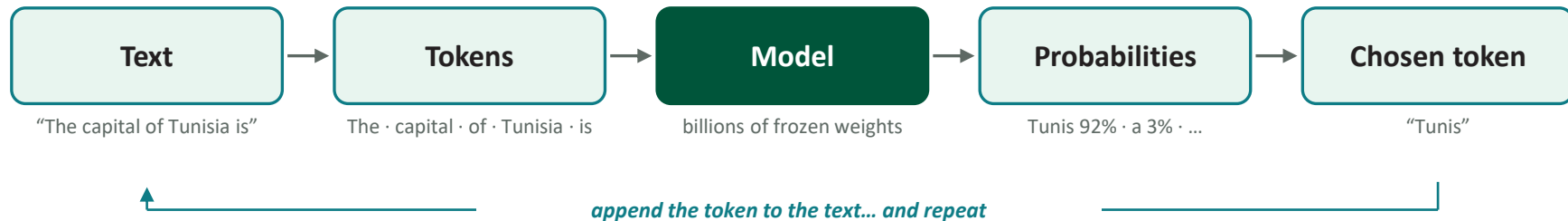
Inference: apply f to a new input — every time it is used

	Linear regression (familiar)	Large language model
Parameters	2 (β_0, β_1)	hundreds of billions of weights
Estimated by	least squares	gradient descent on text
Input $\rightarrow$ output	variables $\rightarrow$ a number	tokens $\rightarrow$ next-token probability
Interpretable?	yes, each coefficient has a meaning	no, in practice

A model is not an application. The chat assistant you use wraps a model in scaffolding: interface, hidden instructions, tools.

G. E. P. Box (1979): “all models are wrong, but some are useful” — LLMs included.

An LLM predicts the next token — that's all it does



Fluency ≠ accuracy

Inside the model there is no database of facts and no notion of truth. A right figure and a wrong figure come from the same computation and look identical on the page.

What an LLM does well — and what it cannot guarantee



Strengths



Draft

a first version of a commentary or note



Translate

between English, French, Arabic, Portuguese



Summarise

an 80-page methodology report



Classify

free text into a classification



Code

R, Python or SQL scripts to be reviewed



Structural limits



Hallucination

invents a plausible figure or reference



No source

cannot say where a learned fact came from



Knowledge cutoff

unaware of anything after its training



Variability

two runs can give two different answers



Context window

can only read a limited amount of text

The model on one side, the scaffolding on the other

THE MODEL

what you buy or download



Changed by: pre-training, fine-tuning

costly · permanent

THE SCAFFOLDING

what you build around it

cheap · reversible

MCP · standard cable to your tools and data

AGENT LOOP · output becomes action

CONTEXT · prompt + retrieved documents

LLM



Nothing on the right changes the model. You can change the scaffolding on a Tuesday afternoon; the model, you cannot. That one fact explains most of the cost and most of the risk.

SECTION 04

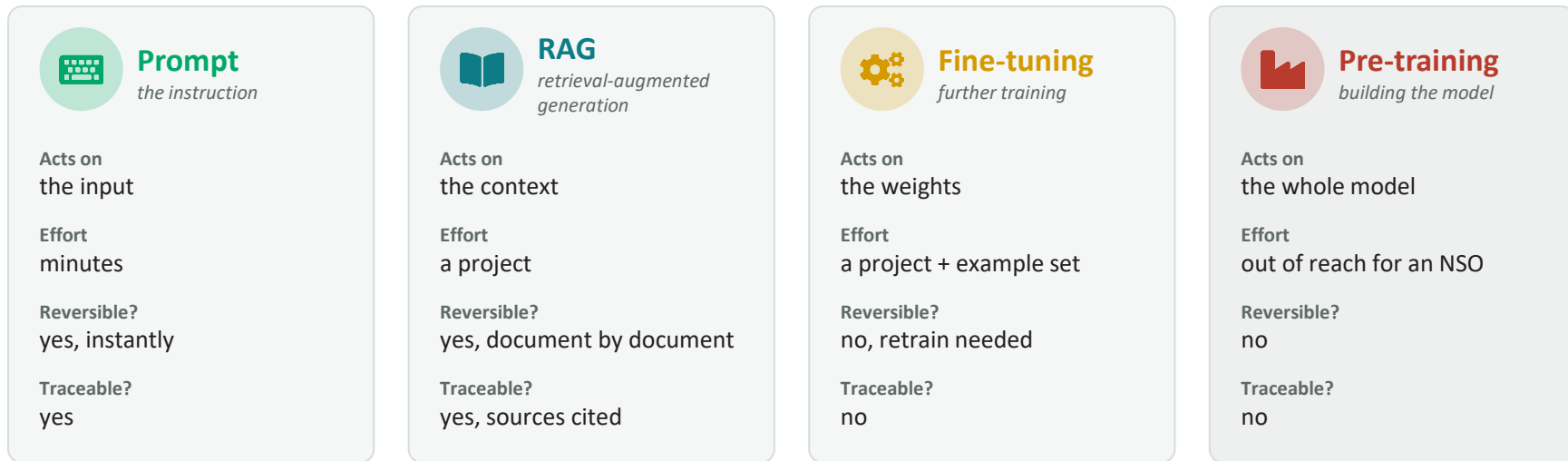
04

Steering: prompt, RAG, fine-tuning

Where do you intervene to change what the model produces?



Four places to change what the model does



cost and permanence increase to the right

 **Rule:** never move right until you have proven the left is not enough. Almost everything is solved on the left.

Prompt engineering: writing an instruction you can check

EXAMPLE OF A STRUCTURED PROMPT

ROLE

You are a writer at the national statistics office.

TASK

Write an 80-word commentary on the table provided.

CONTEXT

[March 2026 CPI table pasted here]

FORMAT

One paragraph, neutral tone, rates rounded to 0.1 point.

CONSTRAINTS

Use only figures from the table. If information is missing, write "not available".



What it changes

The form, tone, structure and scope of the answer.



What it does not change

What the model knows, and its capacity to be wrong.



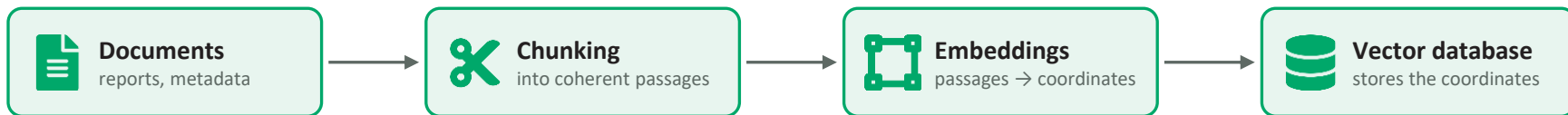
Good practice

Treat prompts like code: versioned, tested, documented.

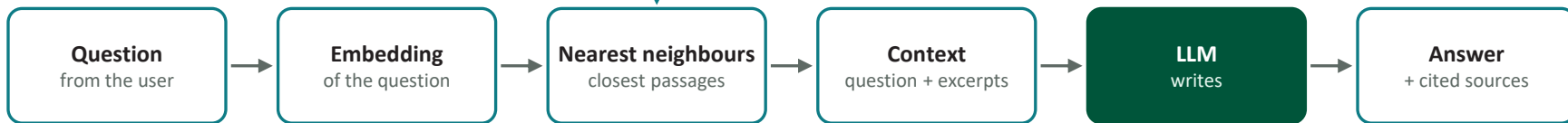
Wei et al. (2022), "Chain-of-Thought Prompting", NeurIPS 35 — how a prompt is structured changes the output.

RAG: retrieve first, then write

PREPARATION — ONCE



FOR EVERY QUESTION



Geometry, not keywords

Passages with the same meaning end up as neighbours: you find a relevant paragraph with no word in common.



Chunking decides

Split a table from its header and no embedding will save you. Quality is often won or lost here.

Lewis et al. (2020), NeurIPS 33 — the founding RAG paper · Muennighoff et al. (2023), MTEB, EACL.

RAG changes what the model sees; fine-tuning, how it speaks



RAG

Changes	the context, at question time
Good for	facts, documents, your data
Traceable	yes — the source comes back with the answer
Updating	replace a document; immediate effect

VS



Fine-tuning

Changes	the weights, once, in advance
Good for	format, register, a repeated task
Traceable	no — nothing points to a source
Updating	retrain with an updated example set



Steering rule

If the requirement says “up to date”, the answer is RAG: weights freeze the day training stops.



Fine-tuning teaches style, not facts

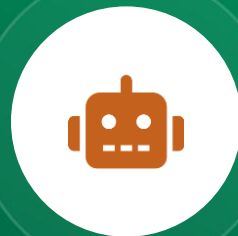
“Fine-tuned on your data” often describes... RAG. Ask in writing whether the weights change.

SECTION 05

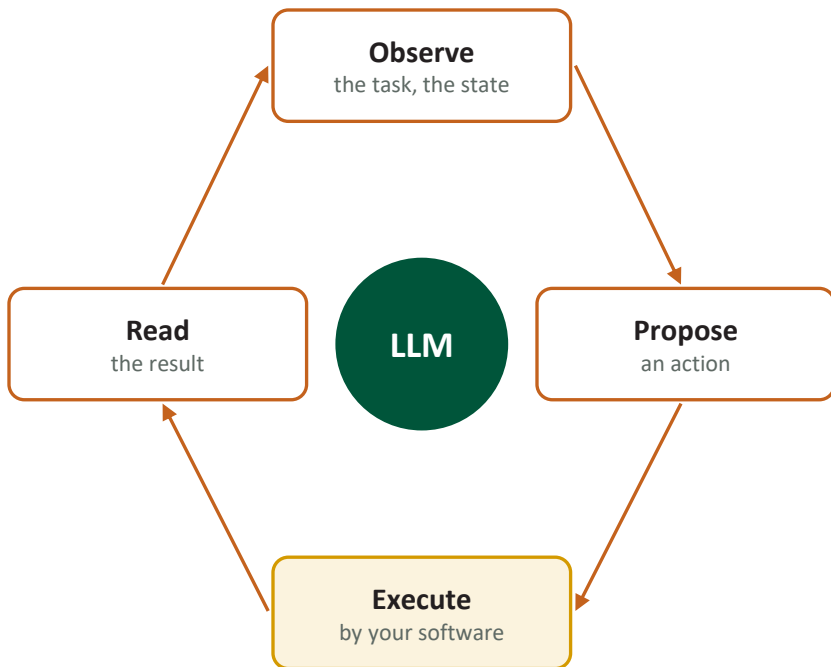
05

Acting: agents, tools and MCP

At what point does an answer become an action?

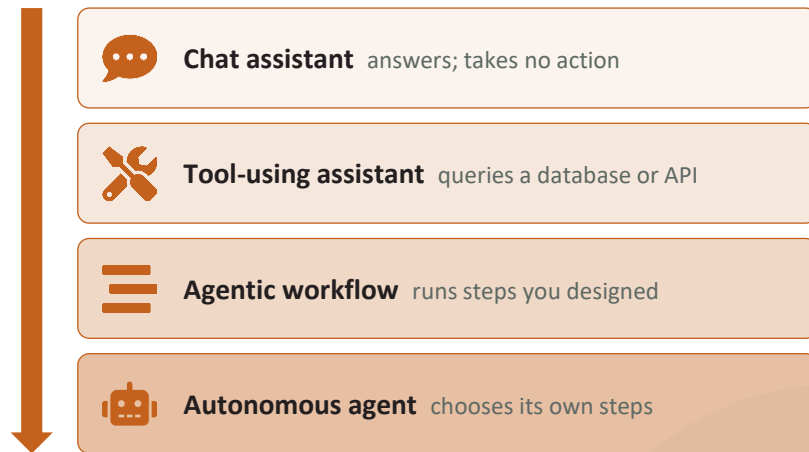


An agent is a model whose output may become action



Yao et al. (2023), “ReAct”, ICLR — the observe, reason, act loop.

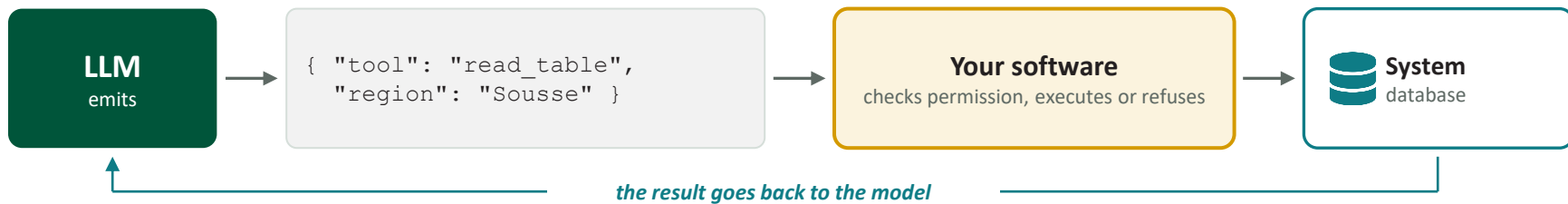
FROM SIMPLEST TO MOST AUTONOMOUS



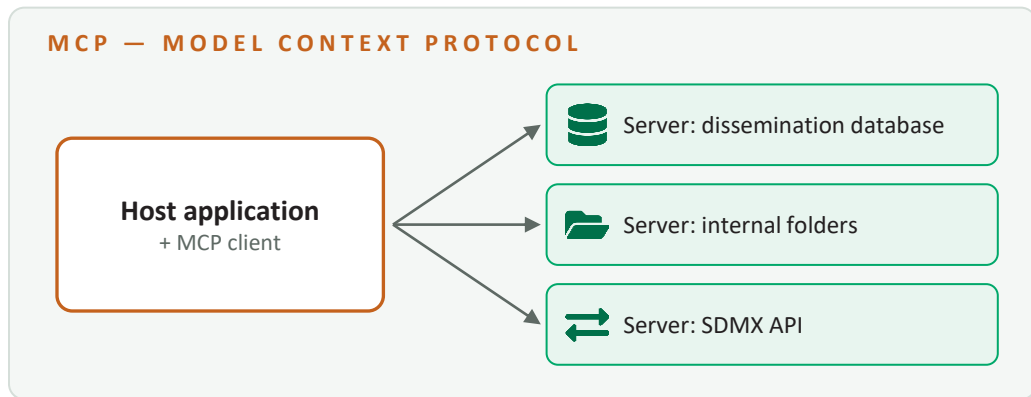
increasing autonomy and risk ↓

“Agentic” names an architecture, not a level of intelligence.

A tool call is a request; MCP is a standard plug



MCP — MODEL CONTEXT PROTOCOL



MCP IN BRIEF

- Open standard published by Anthropic in November 2024.
- Donated in December 2025 to the Agentic AI Foundation (Linux Foundation).
- Current specification: version 2026-07-28.
- **Confers neither intelligence nor authorisation.**

Attack surface = the list of tools you expose. Exposing a tool is an act of governance · modelcontextprotocol.io (accessed Sept. 2026).

SECTION 06

06

For official statistics

What can each concept — and what can't it — do for you?



What each concept can — and cannot — do for you

Concept	Can help you...	Cannot...
● AGI	nothing operational today: it is a research horizon	serve as a procurement or planning criterion
● LLM	draft, translate, summarise, propose code for review	produce a publishable figure or guarantee its source
● Prompt	set the format, tone and limits of a task	add knowledge to the model
● RAG	answer from your reports, citing the source	fix documents that are wrong or badly chunked
● Fine-tuning	learn your house style and conventions	install facts reliably and attributably
● Agent	chain search, query, check and draft steps	decide on its own what gets published
● MCP	connect a model to your systems without custom adapters	authorise, secure, or make anything smarter

Vocabulary discipline: from promise to question

“

“We fine-tuned the model on your data.”



Do the weights change, or is it RAG? In writing.

“

“Our AI doesn't make mistakes.”



What error rate, measured on which test set?

“

“A fully autonomous agentic solution.”



Which tools are exposed, and who authorises each action?

“

“MCP-compatible, so it's secure.”



Which authentication, which audit logs?

“

“We're close to AGI.”



Which measured capability, on which task, today?

“

“Your data stays with you.”



Where does the computation run, under which jurisdiction?

“AI-ready”: readable and reliable for machines



AI-ready: data that are structured, documented and disseminated so that a machine can find, read and interpret them correctly — without losing their meaning.

AI-ready data

five areas from the US Department of Commerce guidance (2025)

-  **Documentation** definitions, methods, units — human- and machine-readable
-  **Data and metadata formats** structured and open: SDMX, CSV, JSON
-  **Storage and dissemination** stable APIs, persistent identifiers and URLs
-  **Licensing and usage** explicit and machine-readable terms
-  **Integrity and quality** revisions, dates, quality indicators

An AI-ready institution

the other half of readiness

-  **Governance**
a written usage policy
-  **Skills**
to evaluate, not just use
-  **Infrastructure**
compute, security, sovereignty
-  **Evaluation**
testing before any adoption



Why it is urgent: general-purpose assistants often give plausible but wrong official figures. AI-ready data let them query your APIs (via RAG or MCP) instead of inventing.

US Department of Commerce (2025), Generative AI and Open Data · FCSM 25-03 (2025), AI-Ready Federal Statistical Data · IMF (2026), StatGPT.

Three layers, eight words: the whole relationship



Inside the model

PRE-TRAINING · FINE-TUNING

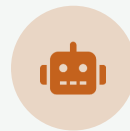
Build and adjust the weights.
Permanent, expensive, unable to cite
a source.



Around the model

PROMPT · RAG

Set the instruction and fill the
context with retrieved documents.
Cheap, reversible, traceable.



Beyond the model

AGENT · AGENTIC · MCP

Let output become action through
standard tools. None makes the
model any smarter.



And AGI? Neither a layer nor a product: a research horizon whose very definition is still disputed. It does not belong in a specification.

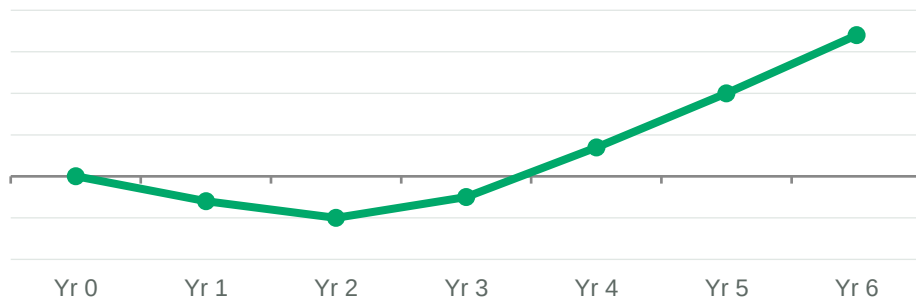
Exposure is broad; the payoff is slow

80%

of US workers have at least 10% of their tasks exposed to LLMs.

Exposure does not mean replacement.

Productivity J-curve (conceptual sketch)



Daily work

A first draft costs almost nothing; the judgement to pick which draft to keep becomes scarce.



Your office

Your staff already use these tools. A policy of silence is a policy of shadow use.



Learning

Gains are largest for novices: assess the process, not just the product.

Eloundou et al. (2024), Science 384 · Brynjolfsson, Rock & Syverson (2021), AEJ: Macro 13(1) · Noy & Zhang (2023); Bick et al. (2024); Brynjolfsson, Li & Raymond (2025).

The red line: no layer produces a published figure

What AI can do

- ✓ draft the commentary
- ✓ retrieve the methodology document
- ✓ propose a code for a staff member to confirm
- ✓ explain a method to a colleague



What stays with the estimator

The published figure comes from an estimator you can **write down, reproduce and defend** to a user who challenges it.

That obligation predates these technologies and is not changed by them — even when an agent would be faster.

“ *A statistic that nobody can reconstruct is not an official statistic, however close to the truth it may be.* ”

United Nations Fundamental Principles of Official Statistics, Principle 2 — General Assembly resolution 68/261 (2014).

WHAT'S NEXT ON DAY 1

At 10:30, you will build the middle column.

RAG is the only setup in this stack that forces a model to show where its answer came from.

 10:30

HANDS-ON LAB

Retrieval-augmented generation
(RAG)

 13:30

TALK

AI infrastructure: what you really
need

 15:00

HANDS-ON LAB

Build an agent, then break it



Bring: a real question from your office that a document could answer.

APPENDIX

A

Held in reserve for questions

Glossary, two cultures, learning families, pre-training cost, sources.



Glossary: the words of the week, in both languages

Model <i>Modèle</i>	a function whose parameters are estimated	AI-ready <i>AI Ready</i>	data readable and reliable for machines
AI <i>IA</i>	a machine performing a task seen as intelligent	AGI <i>AGI</i>	hypothetical human-level AI across varied tasks
LLM <i>GML</i>	large language model; predicts the next token	Token <i>Jeton</i>	piece of a word processed by the model
Prompt <i>Requête</i>	instruction given to the model	Context window <i>Fenêtre de contexte</i>	amount of text readable at once
RAG <i>Génération augmentée</i>	retrieve documents, then write	Embedding <i>Plongement</i>	a passage turned into coordinates
Fine-tuning <i>Affinage</i>	retraining that changes the weights	Hallucination <i>Hallucination</i>	fluent but false output
Agent <i>Agent</i>	model whose output triggers actions	MCP <i>Model Context Protocol</i>	standard for connecting tools and data

Two cultures: explaining the process vs predicting the case



Data modelling

Goal

understand how the data were generated

Validation

goodness of fit, residuals, assumptions

Parameters

have a defensible meaning

Fails when

the assumed model is wrong



Algorithmic modelling

Goal

predict the next, unseen case

Validation

out-of-sample accuracy

Parameters

no direct interpretation

Fails when

tomorrow stops looking like yesterday

Breiman, L. (2001). *Statistical Modeling: The Two Cultures*. *Statistical Science* 16(3), 199–231.

Supervised, unsupervised, reinforcement: three questions



Supervised

Do you have labelled examples of the right answer?

e.g. A census with hand-coded occupations: the codes are the labels.



Unsupervised

Only raw records, and you are looking for structure?

e.g. Deduplicating and clustering the business register.



Reinforcement

A system that acts, and something that scores its actions?

e.g. Almost no NSO use cases today — the most oversold branch.

The same algorithm can serve all three: the question is what you hold, not what you buy.

Why an NSO will not pre-train its own LLM

20 : 1

training tokens per parameter for a
compute-optimal model



An order of magnitude

At the scale of the models used this week, that means a corpus and a compute budget no NSO will assemble — nor should.



Don't confuse them

Pre-training (out of reach) and fine-tuning (accessible, e.g. with LoRA on a single GPU).



The real decision

Choosing, evaluating and governing an existing model — not building one.

Hoffmann et al. (2022), Training Compute-Optimal LLMs, NeurIPS 35 · Hu et al. (2022), LoRA, ICLR · Detrmers et al. (2023), QLoRA, NeurIPS 36.

Sources: where every claim comes from



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Sources: where every claim comes from



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Sources: where every claim comes from



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